

**WHEN KING'S ECOLOGICAL INFERENCE FAILS:
DIAGNOSING THE SMALL-N PROBLEM IN
DATA-SPARSE ENVIRONMENTS USING MOUNT
PLEASANT ELECTION DATA**

BY

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Abstract

This thesis examines the small-N limitation in Gary King’s Ecological Inference (EI) model and explores how data structure and noise affect model convergence in data-sparse environments. Ecological inference is commonly used in racial polarization and Voting Rights Act analysis to estimate individual-level voting behavior from aggregate precinct-level data. While prior literature has acknowledged that small sample sizes can create issues for EI, this thesis argues that model failure is not driven by sample size alone, but rather by the interaction between small-N settings and unstable data distributions.

Using precinct-level demographic and election data from Mount Pleasant, New York, this study evaluates EI performance through three empirical tests: a threshold analysis, a bootstrap analysis, and a simulation study. The threshold analysis tests whether a minimum sample size is required for convergence, the bootstrap analysis evaluates how consistently the model converges under repeated resampling, and the simulation study isolates the effect of noise by testing EI under idealized conditions.

The findings demonstrate that no universal minimum sample size exists for EI convergence. Instead, convergence depends heavily on the underlying structure and stability of the data. The simulation study further shows that EI can perform well even at small sample sizes when noise and instability are minimized. These results suggest that researchers should use caution when applying EI in small-N jurisdictions and should evaluate data stability alongside sample size before interpreting estimates.

Dedicated to my younger self, who had the courage to embark on this challenging journey
(and many others).

I declare that generative artificial intelligence (AI) tools were used in this dissertation as follows:

- **AI Tools Used:** Claude for formatting and word choice. ChatGPT for code debugging and assistance with word choice.
- **Sections/Pages:** Throughout the dissertation, as needed for editing, code assistance, and clarity improvements.
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List of Abbreviations

EI (King's Ecological Inference Software)
RPV Racially Polarized Voting
VRA Voting Rights Act

Chapter 1

Introduction

Patterns of racially polarized voting are the linchpin evidence in Voting Rights Act (VRA) litigation. Because the United States uses a secret ballot, researchers cannot directly observe individual voting behavior, making it difficult to determine whether racially polarized voting (RPV) is occurring in a given jurisdiction. Following the passage of the VRA at the federal level, many states, including New York, have passed their own legislation to remedy systemic issues preventing minorities from voting. However, these laws presume that political scientists have a working methodology to address the RPV linchpin issue, which is not always the case.

To get around the secret ballot problem, enforcement of the VRA requires ecological inference, the process of inferring individual behavior from aggregate data. Since the 1950s, various statistical methods for performing ecological inference have been proposed, the most prominent being Goodman's Regression, the method of bounds, and King's Ecological Inference. Developed in 1953, Goodman's Regression estimates individual behavior by regressing the aggregate output on aggregate input. This is especially liable to produce impossible estimates on voting behavior, and this method assumes that group patterns are consistent across each ecological unit, which has clear methodological flaws (Barreto et al., 2019). Created in 1953, the method of bounds constructs deterministic bounds for ecological inference. These

bounds calculate the minimum and maximum possible values of ecological data. However, while these bounds are helpful, they most often provide wide ranging estimates on group behavior. Finally, King’s Ecological Inference, which is often praised as the “standard” for conducting RPV research, utilizes a maximum likelihood framework that incorporates both aggregate-level regressions and deterministic bounds for each unit (King, 1997). While King’s model has been criticized, it remains the leading model for conducting RPV research and is required under the VRA (de Benedictis-Kessner, 2015).

This paper aims to address two underexplored problems with King’s EI (EI) in the context of RPV research: its unpredictability in data-sparse environments and the issue of noisy data. As RPV cases continue to be brought to courts, the need for a reliable measurement of RPV becomes increasingly important. Without methods that can account for the limitations of small-N data, analyses risk misrepresenting voting behavior, which can have meaningful consequences for the enforcement of voting rights and the broader integrity of the democratic process. In the process of diagnosing and attempting to implement a fix to the turnout assumptions implicit in King’s EI model, I came across an issue that impacts the efficacy of this model: the small-N problem. The small-N problem emerges in data-sparse environments when the data lacks a clear structure. This raises doubts as to whether the model is capable of producing valid estimates. To test if this issue actually exists, I performed three tests: a threshold test, a bootstrap analysis, and a simulation-based analysis. The threshold test analyzes if, when the distribution is held constant and the sample size increases, the model performs differently. A bootstrap analysis of EI convergence across sample sizes determines that EI does not fail uniformly in small-N contexts. Then, a simulation-based analysis of EI performance in data-sparse environments concludes that sample size alone does not determine whether EI converges, but rather the interaction between small-N and data structure. Finally, to visually assess these data characteristics, the relationship between demographic composition and vote share, which EI relies on, is plotted, showing how data structure impacts the success rate of the model.

Ultimately, this paper contributes to ecological inference literature by identifying a specific mechanism of model failure, showing that instability arises from the interaction between small sample size and data structure, rather than from sample size alone. Rather than attributing instability to solely sample size, I show that model failure arises from the interaction between small-N and the underlying data distribution, which produces unstable bounds and prevents the model’s numerical integration from converging. This provides a clearer explanation for and when and why EI fails in applied settings relevant to VRA litigation.

In this study, Mount Pleasant data was used because the jurisdiction was recently litigated for violations of the VRA, and I aimed to assess whether King’s model could detect RPV in this dataset. Additionally, because the data have a small N and lack a clear data distribution, they proved useful for this analysis.

Mount Pleasant is particularly interesting as a case study because it falls within New York State’s school district governance structure. Unlike other states where school boards are appointed, New York has direct elections. This makes school board elections potential sites for RPV to occur (Magleby, 2026). This creates a vote dilution concern that the VRA was intended to address: when certain populations are systematically denied access to educational resources, inequality is perpetuated. However, empirical analysis of NY school district elections is limited, partially because of methodological challenges that this thesis presents. Because school districts often have few precincts, the small-N problem emerges, making standard EI difficult, or even impossible, to apply.

This thesis is organized as follows. First, I will discuss the theoretical framework of this work, beginning with the legal basis, then RPV theory, followed by Ecological Inference, and ending with its key limitations. Then, I will discuss my methods and research design, consisting of threshold test of sample size, a bootstrap analysis of EI convergence across sample sizes, and a simulation-based analysis of EI performance in data-sparse environments. Third, I move into my findings. In this section, I review the results of these tests and

visualize election data to see why EI model may be breaking. Finally, I conclude this work by discussing why EI must be improved to better handle data-sparse jurisdictions and present directions for future research.

Chapter 2

Theoretical Framework

The question of whether minority voters have equal opportunity to participate in the political process and elect representatives of their choice is an essential part of the American voting rights framework. Since the passage of the VRA of 1965 and the Supreme Court’s landmark decision in *Thornburg v. Gingles* (1986), courts and scholars have relied on empirical evidence of RPV. RPV is the phenomenon where voters of different racial groups systematically prefer different candidates. This evidence is used to identify and remedy structures that disadvantage minority communities. Establishing the presence (or absence) of RPV requires inferring individual-level behavior from aggregate precinct data. This is referred to as the ecological inference problem. Since Gary King’s (1997) seminal work, ecological inference has become the standard empirical tool for performing RPV analysis in both academic research and VRA litigation. However, this method has many well-documented limitations, including sensitivity to small samples sizes, and an implicit turnout assumption. While RPV has been studied in large jurisdictions, small suburban cities have been comparatively understudied. This thesis addresses that gap by analyzing RPV in Mount Pleasant, NY, using ecological inference.

2.1 Legal Framework

Before the passage of the VRA (1965), minority voters faced serious systemic barriers barring them from their right to vote, such as poll taxes and literacy tests as a prerequisite to voting. The passage of the VRA outlawed these discriminatory voting practices. According to the National Archives, this bill was "the most significant statutory change" to the relationship between state and federal government since the Reconstruction Era following the Civil War. This Act had an immediate impact: by the end of 1965, 250,000 new Black voters had been registered to vote (National Archives, 1965).

The most consequential component of the VRA for litigation is Section 2:

No voting qualification or prerequisite to voting, or standard, practice, or procedure shall be imposed or applied by any State or political subdivision to deny or abridge the right of any citizen of the United States to vote on account of race or color.

This clause establishes that no state or local government may impose voting rules that discriminate based on race.

While the period between 1965 and 1982 saw ongoing litigation and enforcement of the VRA, the core legal shift relevant to this analysis occurs in the 1982 amendments, which moved the focus from intent to the effect of RPV.

A landmark case that operationalized Section 2 of the Voting Rights Act is *Thornburg v. Gingles* (1986), which allowed minority groups to have a level of political participation they were previously denied. This case set the "discriminatory effect precedent", which is used in deciding whether legislative maps violate the law when there is evidence of racial vote dilution in redistricting methods. This case established a 3-prong test to determine whether a district was non-compliant with the VRA.

The three necessary preconditions for a Section 2 claim are as follows (Thornburg v. Gingles, 1986):

- Compactness: the minority group must be large and geographically compact enough to be a majority in a single-member district.
- Cohesiveness: the minority group must demonstrate that its members tend to vote as a bloc.
- Majority bloc voting: the minority must show that the white majority votes as a bloc that usually defeats the minority’s preferred candidate.

This case provided the opportunity for minority voters to use evidence to combat systemic structures that diluted their votes, allowing them to elect representatives who were more responsive to their needs (Hagen, 2024).

2.2 Racially Polarized Voting Theory

In the context of voting rights litigation, RPV occurs when the political preferences of the majority-race and minority-race voters diverge substantially. In this case, substantially means that majority-race voters have enough cohesion to consistently defeat the minority-race’s preferred candidate.

In order to qualify as RPV, two conditions must hold: preference polarization and voting power. Political polarization is necessary for RPV because RPV is defined by systematic differences in candidate preferences across racial groups. Without these differences, voting behavior is not racially polarized. Voting power is a necessary condition because RPV only matters if the difference in preferences between the minority and majority groups can actually change the outcome of an election.

Determining whether RPV exists in a given jurisdiction is methodologically challenging. This is because demographic information of voters is not included on ballots. Elmendorf et al. explains “[voting] preferences must be estimated, and the estimation of candidates’ vote shares by racial group from ballots cast in actual elections depends on strong assumptions about political homogeneity within racial groups across geographic areas” (Elmendorf et al., 2016). The Yale Law Journal (Chen and Stephanopoulos, 2021) outlines a method for

defining districts that have RPV by using statistical estimates of voting behavior to determine where minority voters have the power to elect their preferred candidates. They performed the following procedure to establish if RPV exists in a district. First, the authors utilize ecological inference software to estimate minority cohesion and the extent to which white voters vote as a bloc to consistently defeat the minority-preferred candidate. Next, they define opportunity districts: districts in which the minority-preferred candidate wins the general election. In these districts the number of minority voters supporting a candidate outweighs the number of white voters supporting that candidate. This measures the political power of the minority voters in a district. Third, the authors scale this analysis across 19 states, and use the 2012 presidential election as a proxy for measuring RPV, as precinct-level data was generally available for this election, and it featured a candidate (Barack Obama) who was clearly minority-preferred, making it a good choice for estimating RPV patterns. Finally, the authors evaluate whether RPV outcomes for existing district boundaries are better or worse than simulated boundaries. Currently, RPV law is applied to see if minority representation is proportional to the population. However, the authors apply RPV to their "race-blind baseline" to see if the current level of minority representation is higher than what would 'naturally' occur if a computer drew the lines without any racial data. Ultimately, the authors determine that mapmakers have used RPV data to 'pack' minority voters into districts beyond what is necessary for them to be able to elect their candidates of choice. This results in districts that are more minority-heavy than those in race-blind simulations. Chen and Stephanopoulos show that RPV is not just a theoretical concept, it's a means of defining minority opportunity districts and determining who has political power.

While RPV analysis is helpful in determining districts that may be violating Section 2 of the VRA, it comes with its own complications and nuances. These complications are outlined in *Reimagining Democratic Inclusion: Asian Americans and the Voting Rights Act* (Chen and Lee, 2013). This article applies the concept of RPV and argues that traditional methods for measuring RPV are no longer viable in modern, multiracial electorates. Their

main argument is that RPV treats RPV as a "Black-White Paradigm" that is not inclusive of other minority groups. This paradigm is rooted in the historical struggle of African Americans, and creates a faux dichotomy, when a myriad of other racial groups exist. The authors go on to explain why many statistical models, such as Goodman's Regression and the method of bounds, are not well-suited for cases with more than two racial groups. As the number of groups increases, so do the number of unknowns, making it much harder to reliably estimate individual voting behavior from aggregate data. Finally, the article reimagines RPV for Asian Americans and proposes a reform to RPV that includes more cultural competency. This view incorporates "cultural compactness" and "communities of common interest" as opposed to solely relying on bloc voting. Courts should look at additional data of the attitude of voters such as whether minority voters believe what happens to other racial groups affects their own lives, as well as shared language, economic status, and parallel experiences of discrimination. Ultimately, the authors argue that scholars should relax the traditional Gingles RPV requirements, which would allow for the protection of minority communities even when they do not all vote as a bloc.

While RPV analysis grows more complex in multiracial contexts, this study adopts the traditional two-group Gingles framework comparing minority and white voters. I define two categories - white and minority voters. Minority voters, including Hispanic, Black, Asian, and Pacific Islander groups, aggregate individuals across racial categories that may not share uniform political preferences. As scholars have noted, RPV analysis in multiracial contexts risks hiding meaningful variation within the minority community itself. This limitation is acknowledged, and findings should be interpreted accordingly.

2.3 Ecological Inference Methodology

Performing RPV analysis requires the challenge of inferring individual behavior from aggregate data. This task is known as the ecological inference problem. When individual votes

are aggregated to the precinct level, we lose information about individual-level behavior and the relationship between group identity and vote choice. This makes it difficult to determine how different groups actually voted. This could lead to ecological fallacy, a term coined in 1950 by William S. Robinson to describe a logical error in data analysis where inferences about the nature of individuals are wrongly deduced from inference for the group of which they belong. For instance, a scholar who commits ecological fallacy might assume that if Black voters appear to vote Democrat in a precinct, then any individual Black voter in that precinct voted Democrat. This practice is called a fallacy because it is based on the problematic assumption that relationships that hold on the aggregate level also hold on the individual level (de Munck, 2005).

In his seminal work, Dr. Gary King proposes a solution to the ecological inference problem in his eloquently named work *A Solution to the Ecological Inference Problem: Reconstructing Individual Behavior from Aggregate Data*. In this work, King outlines a 7-characteristic solution to the ecological inference problem, and details why previous methods of combating this problem are inadequate. King is careful to outline that his book focuses on *a* solution to the EI problem, not necessarily *the* solution.

Briefly, the 7 characteristics of King’s solution are as follows:

1. This solution is scientifically validated with real data
2. This method offers realistic assessments of the uncertainty of ecological estimates
3. The basic model is robust to aggregation bias
4. All components of this proposed model are, in large part, verifiable in aggregate data
5. This solution corrects for a variety of serious statistical problems, unrelated to aggregation bias, that also affect ecological inferences, like the indeterminacy problem and grouping problem
6. This method provides accurate estimates of voting behavior for both overall aggregates (like district and state) and for individual precincts
7. The solution to the EI problems becomes a solution to what geographers’ call a ‘modifiable areal unit problem,’ or a source of statistical bias in spatial analysis where aggregating point data into arbitrary zones can yield wildly different results based on the shape and size of those zones

When the work was written in 1997, only Goodman’s regression was utilized to solve this problem. Goodman’s regression is based on straightforward linear regression and effectively assumes that quantities of interest are constant over precincts. King’s model doesn’t make this assumption, which King believes moderately improves results. Compared to other statistical solutions, the major improvement that King’s model boasts is that it produces deterministic bounds. These bounds set rigid constraints on the possible range of values of group voting behavior, reducing the uncertainty of statistical estimation. These bounds are derived from known constraints such as voter turnout percentages, and limit the possible solutions to what is feasible by the data.

In his follow-up work, King collaborates with Ori Rosen and Martin Tanner to identify the 3 core assumptions of King’s model: that parameters follow a truncated bivariate normal distribution, parameters do not have aggregation bias, and that data exhibits no spatial autocorrelation (King et al., 2004).

While King’s contributions have shaped how RPV analysis is conducted, his methods have faced significant scholarly critique, particularly regarding the assumptions embedded in his statistical model. For instance, Jon Wakefield argues that the ecological inference problem is fundamentally non-identifiable, meaning that many different internal cell counts can produce the same aggregate totals (Wakefield, 2004). This means that the true underlying voting patterns cannot be determined from the data alone, making uncertainty a fundamental part of EI. In terms of RPV, this means that different combinations of minority and white voting behavior could all be consistent with precinct-level results. This makes EI estimates heavily model-dependent, and means that the assumptions the model relies on are central to the validity of RPV analysis. This issue is especially apparent in data-sparse environments, as overfitting becomes problematic. Additionally, Wakefield critiques the validity of King’s diagnostic tools. Ultimately, Wakefield concludes that the most robust solution to the ecological inference problem is combining aggregate data with individual survey data. This adds more information to the data, and he finds that even a very small individual-level

sample can dramatically reduce bias and add precision in a way that aggregate data cannot provide on its own. While incorporating this level of modeling is outside the scope of this thesis, it provides an interesting direction for future research.

One of the most well-known critiques of King’s EI comes from David A. Freedman and colleagues, who raises concerns about the method’s performance, accuracy, and the reliability of its diagnostic tools. For example, when tested on datasets where individual-level voting behavior is actually known, King’s method produced estimates that were significantly off. Freedman et al. argue that the diagnostic tools built into King’s model do not reliably detect when the model is misspecified. These tools are subjective and do not provide clear guidelines for when a model should be rejected, which makes them harder to rely on. In addition, Freedman points back to the core issue of non-identifiability in ecological inference, emphasizing that multiple different combinations of individual behavior can produce the same aggregate data. He ends with his strongest critique, arguing that calling EI a “solution” to the ecological inference problem is misleading as the method does not consistently outperform existing approaches (Freedman et al., 1998).

Shortly after Freedman’s article was published, Gary King responded, defending his approach. He stated that the strength of his EI method comes from combining three types of information that earlier methods typically used separately. First, the model uses the method of bounds to establish the range of possible values for precinct-level quantities. Second, it borrows strength across geographic units to narrow these ranges into a joint posterior distribution. Third, it includes diagnostic tools that are meant to help researchers assess whether the model’s assumptions are reasonable. King emphasizes that ecological inference is not just a purely mathematical exercise, but a framework that allows researchers to incorporate outside knowledge, such as historical context, demographic patterns, and qualitative insights, into the analysis (King, 1999). This flexibility makes the model adaptable.

Overall, this exchange shows that there is still debate about how well King’s EI method performs, especially in terms of the assumptions built into the model. Despite these critiques,

the flexibility of King’s framework and the structure it provides make it a useful starting point in RPV analysis. Because of this, I use King’s EI model as the foundation for my analysis, while focusing on how it performs under small-N conditions and varying data structures.

2.4 Limitations

With this broader debate in mind, I now focus on the limitations of ecological inference that are most relevant to my analysis, particularly those related to the turnout assumption and the issue of small sample sizes with EI.

2.4.1 The Turnout Assumption

In King’s model, the turnout assumption is implicit. The equal turnout assumption states that group voter turnout is assumed to be constant across precincts, regardless of demographic composition. In his work *Aggregation*, Voss defines the turnout assumption as a primary pitfall in ecological inference, and notes that analysts, like King, often mistakenly assume that minority turnout rates are constant across precincts, regardless of differences in demographic composition (Voss, 2005). This assumption allows analysts to bypass modeling turnout and move directly to an analysis of vote choice. While, in a perfect world, this assumption may hold, in reality, factors such as income, education, and cultural and geographic norms influence individuals’ likelihood of turning out to vote. The consequences of this assumption can be severe, as it introduces systematic bias into estimates. Voting is a complex process involving registration, turnout, and vote choice, and treating it as a “black box” by ignoring these distinct stages increases the likelihood of producing unreliable and poorly understood estimates. If minority voters turn out at lower rates than white voters but equal turnout is assumed, the model will incorrectly attribute differences in turnout to differences in vote choice, leading to biased estimates of group-level voting behavior. King attempts to address this problem by utilizing the method of bounds. However, while these

bounds can be constructed, the variability of estimates within them can still result in unreliable conclusions.

2.4.2 Small Sample Limitation

Jon Wakefield also offers a more technical analysis of EI in his work *Ecological Inference for 2×2 Tables*, where he examines how sparse data makes statistical outcomes highly dependent on population size and the choice of prior distributions. He argues that posterior results are often very sensitive to prior specifications, and when N is small, there often isn't enough data to "cover up" the influence of non-informative priors. He goes as far to describe non-informative priors as "dangerous" in the context of EI. For example, using the standard $\text{Gamma}(0.001, 0.001)$ prior can produce a distribution on cell probabilities that is heavily concentrated in spikes at the boundaries of 0 or 1, creating a U-shape that often doesn't reflect reality. Ultimately, Wakefield concludes that there is no safe choice of non-informative priors in EI, and advocates for incorporating substantive prior information, derived from external data, to better assess how prior assumptions interact with the strength of the observed data (Wakefield, 2004).

In the work *Sensitivity Analyses for Sparse-Data Problems - Using Weakly Informative Bayesian Priors*, Hamra et al. describes the issue of non-informative priors within the Bayesian framework. While this paper analyzes the Bayesian framework in the context of epidemiology research, it provides insight into the small- N problem in EI, as EI utilizes the Bayesian framework. They write, "as the sample size of the study increases, a weakly informative prior will have vanishing impact on model estimates. Specifically, as data becomes less sparse, we would obtain approximately the same point and interval estimates with or without a weakly informative prior" (Hamra et al., 2013). Where data is sparse, which is increasingly likely in small- N situations, prior distributions play a much larger role in shaping estimates. Thus, in small samples, EI estimates are not just unstable, but also more dependent on modeling assumptions than purely the data.

Chapter 3

Methods & Data

3.1 Research Design

In this paper, I investigate the challenges of applying EI to small-sample contexts, using Mount Pleasant precinct data as a case study. Ecological inference is “the process of extracting clues about individual behavior from information reported at the group or aggregate level” (King et al., 2004). When studying precincts where RPV may be present, researchers typically only observe precinct-level election results. By applying EI methods, researchers can estimate how different groups within a precinct may have voted and assess whether RPV may be occurring.

My initial research design aimed to address a known limitation of EI: the implicit assumption that turnout is constant across precincts despite differences in demographic composition. I planned to relax this assumption by incorporating race-specific turnout data that I obtained for Mount Pleasant precincts. However, while attempting to obtain baseline estimates, I encountered a more fundamental issue that has received limited attention in the EI literature: small sample sizes.

I chose to use Mount Pleasant data because this jurisdiction was recently involved in VRA compliance litigation, making it a relevant case for examining whether RPV is

present. In contexts like Mount Pleasant, where the number of precincts is limited ($N = 25$), traditional EI methods faced estimation difficulties. Small sample sizes can lead to unstable parameter estimates, wide confidence intervals, and model convergence issues. These are all problems I encountered repeatedly when attempting to run the standard EI model on my data. This realization led me to pivot my research focus from the turnout assumption to the small-N problem, which represents an important but underexplored limitation of EI methods in contexts where precinct-level data is scarce.

In this research, I will use precinct-level voting data. Ecological inference works by using aggregate data from many geographic units, such as precincts, to infer individual-level behavior. For each precinct, we can observe things like the proportion of minority voters (X_i), or the vote share for a candidate (T_i). We are not able to directly observe how minority or white voters voted within each precinct. This model leverages variation in demographic composition across precincts to estimate these unobserved voting patterns. Precincts are a good unit of analysis because they are relatively small geographic units where voters are more likely to be similar, which reduces aggregation bias as compared to county- or state-level analysis.

However, a strength of EI, which is its reliance on variation across many units, becomes a weakness when the number of available units is small. To explore this limitation, I first attempt to obtain baseline estimates using King’s standard EI model on Mount Pleasant data. This process revealed the extent of the small-N problem and motivated my investigation into potential solutions.

3.2 King’s Ecological Inference Model

EI estimates individual-level data from aggregate data by using a combination of deterministic bounds and a probabilistic model. It attempts to solve the issue of ecological fallacy by modeling the underlying distribution of individual behavior instead of relying solely on

linear regression (King, 1997).

The base formula for EI shows how the observed total vote share in each precinct (T_i) is composed of the voting behavior of different demographic groups. This is an accounting identity, and is also known as the tomography line equation.

$$T_i = X_i \cdot \beta_i^B + (1 - X_i) \cdot \beta_i^W$$

- i : each precinct
- T_i : observed total vote share
- X_i : proportion of minority voters in precinct i , while $1 - X_i$ represents the proportion of white voters
- β_i^B : proportion of minority voters who supported the candidate of interest
 - Vote choice rate among minority voters who actually turned out:

$$\beta_i^B = \frac{\text{Minority votes for the candidate}}{\text{Total minority voters who turned out}}$$

- β_i^W : proportion of white voters who supported the candidate of interest
 - Similarly, the white vote is:

$$\beta_i^W = \frac{\text{White votes for the candidate}}{\text{Total white voters who turned out}}$$

In this formula, β values are not directly observable, as individual vote choices are secret and election results are only reported in their aggregate form. Therefore, this equation shows that the total vote share observed is a weighted average of the voting behavior of each group, where weights are the proportions of a group's population out of the total population. Because β_i^B and β_i^W are unknown, the goal of EI is to estimate these quantities using the observed values of T_i and X_i across many precincts.

Next, EI utilizes the method of bounds to establish the range of possible values for each group's vote share, given the precinct's demographic composition and total vote share. In this model, Duncan-Davis bounds are used.

The Duncan-Davis bounds are determined by the following formulas:

$$\beta_i^B \in \left[\max \left(0, \frac{T_i - (1 - X_i)}{X_i} \right), \min \left(1, \frac{T_i}{X_i} \right) \right]$$

$$\beta_i^W \in \left[\max \left(0, \frac{T_i - X_i}{1 - X_i} \right), \min \left(1, \frac{T_i}{1 - X_i} \right) \right]$$

These bounds constrain the estimates to logically possible values: voting rates must fall between 0 and 1, and the group-specific rates must aggregate to produce the observed precinct-level vote share in T_i .

However, these bounds become less informative in more integrated precincts. When X_i is closer to 0.5, the bounds can be quite wide, allowing for many possible combinations of group-level voting behavior that are consistent with the observed data. In contrast, in more homogeneous precincts, where X_i is close to 0 or 1, the bounds are much tighter, as the observed vote share can be attributed more directly to a single group.

This limitation is consequential for VRA litigation: districts that face legal challenges for potential violations of the VRA often include heterogeneous precincts, where this ambiguity is greatest. Understanding the limitations of EI in such contexts is therefore necessary for assessing the validity and reliability of the voting behavior estimates that inform legal proceedings.

Then, King models the distribution of individual-level behaviors using the truncated bivariate normal distribution. This prior prevents impossible estimates by restricting values to a specific range. In this case, the range is restricted to the values determined by the Duncan-Davis bounds. This prevents impossible estimates that were common in earlier regression models.

King's model then makes the assumption that while precincts vary in demography and vote share, they follow a shared distribution. This allows the model to 'borrow strength' from other observations to provide more precise estimates. This shows that using more data

in the analysis can lead to more reliable estimates, though this depends on the underlying data structure.

Finally, the model performs maximum likelihood estimation and calculates the parameters of this distribution. This outputs estimated percentages of each group, as well as visualizations of these values.

3.3 Data Sources

3.3.1 Dave’s Redistricting App

I collected precinct-level demographic and election data for Mount Pleasant from Dave’s Redistricting App (DRA). DRA is an online platform designed to provide accessible redistricting and demographic data tools to researchers, policymakers, and the public. Its goal is to make redistricting data more transparent and easier to analyze (Dave’s Redistricting, 2020).

DRA provides precinct-level election and demographic data through an interactive mapping platform. This makes it easy for researchers to analyze electoral geography and demographic patterns.

For this project, DRA provided precinct-level election results and demographic estimates in a downloadable format, which made it a good source of data for performing ecological inference.

Data Collection

To collect data from Dave’s Redistricting App, I created a map within DRA filtered by state, shapefile (to determine what the shapes of precincts were), race, city, census year, voting age, and election.

To collect data for the 2020 Presidential Election in Mount Pleasant, I used the following parameters:

- **State:** New York
- **Shapes:** 2020 Precincts
- **Plan Type:** Congress
- **Restrict To:** City of Mount Pleasant
- **Census:** Total Population 2020
- **Voting Age:** Voting Age Pop 2020
- **Election:** President 2020

From there, I downloaded the Precinct Data of my 'Map' as a .csv that can be read by R. This downloads all census, voting age, and election data, by city.

Cleaning and Pre-Processing Data

From the data downloaded, the data columns that are relevant to this research is as follows:

- **V_20_VAP_Total:** Total voting-age population in Mount Pleasant in 2020.
- **V_20_VAP_White, V_20_VAP_Hispanic, V_20_VAP_Black, V_20_VAP_Asian, V_20_VAP_Native, V_20_VAP_Pacific:** Voting-age population for each racial group.
- **E_20_PRES_Total:** Total number of Presidential votes cast in 2020.
- **E_20_PRES_Dem, E_20_PRES_Rep:** Votes cast for the Democratic and Republican candidates.

All other columns were removed during pre-processing. Removed columns contained precinct and district identifiers (such as names or unique IDs), columns describing total population demographics rather than Voting Age Population (VAP) demographics, and columns containing composite results from multiple elections, since this analysis focuses on individual election results.

From here, I have the necessary information to process this data into a format ready to be inputted into EI.

3.4 Variables

EI requires the following inputs:

- X_i : $X_i = \frac{GroupProportion_i}{TotalProportion_i}$. Our demographic group of interest will be the minority group.
- T_i : $T_i = \frac{Votesforcandidate_i}{Totalvotes_i}$. Our candidate of interest will be the Democratic candidate, who won in this district.
- N_i : turnout: the total VAP.

Calculating Relevant Variables

To calculate X_i , the demographic group of interest is minority voters, I first took the summation of the minority voting-age population columns (V_20_VAP_Hispanic, V_20_VAP_Black, V_20_VAP_Asian, V_20_VAP_Native, V_20_VAP_Pacific) to compute the total minority voting-age population. I denote this column as `minority_VAP`. I then calculated

$$X_i = \frac{\text{minority_VAP}}{\text{V_20_VAP_Total}}.$$

To calculate T_i , I treated the Democratic candidate as the candidate of interest and computed

$$T_i = \frac{\text{E_20_PRES_Dem}}{\text{E_20_PRES_Total}}.$$

Finally, N_i represents the total voting age population in each precinct, which corresponds to `V_20_VAP_Total`.

Now that I have the relevant X_i , T_i , and N_i columns, the remaining data is irrelevant and deleted.

3.5 Analytical Approach

Originally, the scope of this research focused on examining the turnout assumption implicitly written in EI. However, attempting to generate baseline EI estimates as an initial step to

modifying the model to account for turnout, a more immediate issue emerged: the small-N problem. This limitation presented a significant barrier to conducting meaningful analysis, and ultimately prompted a shift in focus from the turnout assumption to the implications of small sample sizes in EI.

After pinpointing the small sample size limitation of EI, I performed an investigation on why this limitation occurred. This issue was identified through an analysis of the error message produced by R after King’s standard EI model failed to run. This investigation was conducted through three experiments: a threshold test, bootstrap analysis, and simulated data experiment.

Test	What it Tests	What it Isolates
Threshold Test	Identifies whether a minimum sample size is required for EI to produce estimates.	Sample size is varied while holding the underlying data distribution constant.
Bootstrap Analysis	Evaluates how reliably the model converges across repeated samples drawn from the same dataset.	Sampling variability and data distribution are introduced while controlling for sample size.
Simulation Test	Assesses model performance under ideal, noise-free conditions with known underlying parameters.	Removes real-world noise to isolate the effect of small sample size.

Table 3.1: Summary of Analytical Approach Tests

Together, these tests are designed to isolate whether EI failure is driven by sample size alone or by the interaction between sample size and data structure.

3.5.1 Threshold Test of Sample Size

To diagnose whether small sample size was actually causing this error, I duplicated the 25-precinct dataset from the 2020 Mount Pleasant presidential election to artificially increase N . When I duplicated the dataset once ($N = 50$) and reran the model, the same error occurred. But when the dataset was tripled (so $N = 75$), the model worked. This experiment provides evidence that sample size contributes to model failure, but the later comparison with other datasets shows that sample size alone does not fully explain the problem.

Then, data was iteratively removed from the triplicated dataset to find the 'breaking point' where the model would no longer run. This process determined that the model failed at 51 data points - anything less and it wouldn't run.

To determine whether this issue was driven purely by small-n, rather than by noise in the data, this experiment was repeated using both a sample dataset included in King's EI package and Mount Pleasant's 2024 presidential election data. The sample dataset was restricted to the first 25 observations, while the 2024 Mount Pleasant dataset also contains 25 precincts. Holding the sample size constant at 25 allows for a comparison that isolates differences in the underlying noise of each dataset.

3.5.2 Bootstrap Analysis of EI Convergence Across Sample Sizes

Following the threshold test, I conducted a bootstrap analysis to assess the stability of EI estimates at different sample sizes. Bootstrapping is a resampling technique that allows researchers to estimate properties of a distribution by repeatedly drawing samples, with replacement, from the original dataset (GeeksforGeeks, 2021). This allows us to evaluate whether EI runs reliably across many slightly variable versions of the same dataset. By repeatedly sampling 25-sample datasets from each dataset, with replacement, I generated new datasets with similar distributions to examine whether EI performance varies across datasets beyond differences in sample size. For each bootstrap sample, I attempted to run EI and recorded whether the model successfully converged. The success rate is defined as the proportion of bootstrap samples for which EI returned estimates. These results can be interpreted as an empirical measure of the model's reliability under a given data distribution.

Altogether, the model generated 200 25-sample datasets, drawn from each elections datasets.

3.5.3 Simulation-Based Analysis of EI Performance in Data-Sparse Environments

To evaluate the performance of EI with different amounts of data, I generated a synthetic precinct-level dataset with known underlying voting behavior. This isolates the small-N problem within EI, as the noise present in real-world data, which arises from variation in individual voting behavior, is not incorporated into this simulation. As a result, this represents a best-case scenario for EI performance.

Using this synthetic population, I repeatedly sampled precincts at varying sample sizes ($N = 25, 40, 50, 60, 75, 100$) and estimated EI on each sample. For each run, this experiment tested whether or not EI successfully returned estimates, and, when the model ran, compared the estimated group voting rates to the true values in data generation.

The simulated model was created by generating the following columns for 1000 precincts:

- Precinct ID: numbered from 1 to the total number of synthetic precincts
- Total population: randomly generated for each precinct using a uniform distribution. Uniform distribution ensures that the simulated data does not impose any additional structure (like clustering or skewness) that could potentially influence the results
- Number of White Voters: randomly generated as a proportion of the total population of the precinct
- Number of Minority Voters: `Total population - Number of White Voters`
- Total Votes: `turnoutrate * Total population`
- Number of Minority Votes: `minority democrat rate * Number of Minority Voters`
- Number of White Votes: `Total Votes - Number of Minority Voters`
- Number of Democrat Votes: `(Minority Democrat rate * Number of Minority Voters) + (White Democrat rate * Number of White Voters)`
- Number of Republican Votes: `Total Votes - Number of Democrat Votes`

Then, the X, T, and N values needed for EI were created using the methodology above.

To generate data, I fixed turnout and group voting rates to isolate the effect of sample size and data structure on model performance. Specifically, turnout was held constant across

groups at 0.60, while the white and minority Democratic voting rates were set to 0.30 and 0.90, respectively.

Parameter	Value
Turnout Rate	0.60
White Democrat Rate	0.30
Minority Democrat Rate	0.90

Table 3.2: Simulation Parameter Values

These values were chosen to create a clear and stable signal of RPV. By specifying a large gap between group voting behavior, the simulation ensures that any estimation error or model failure is driven by the properties of the data (such as small sample size or distributional instability) rather than the ambiguity in the underlying voting patterns.

While these assumptions may not fully reflect reality, they simplify the model by isolating the model's estimates, allowing for a clearer comparison between the model's outputs and the actual voter breakdown.

As the model iterated across 200 bootstrapped samples for each size, it tracked whether or not the EI model successfully converged.

Finally, for the data that converged, the model records the following metrics for the sampled data: average success rate (the proportion of samples in which EI successfully ran), the mean estimated minority voting rate, and the mean absolute error (MAE) for each. The MAE shows, on average, how inaccurate the model's predictions are from actual values (Waples, 2025).

Chapter 4

Findings

This section presents the results of the analyses. I first examine the threshold test, which identifies the minimum sample sizes required for EI to produce estimates. I then analyze bootstrap results from the 2020 to 2024 Mount Pleasant Presidential Election datasets to assess how sample size and data distribution affect model convergence. Finally, I evaluate EI performance using simulated data with known parameters to isolate the impact of small sample size under ideal conditions.

The results support a central finding: EI failure is not uniform in small-N settings, but depends on how sample size interacts with the structure of the data. Across all tests, model convergence is determined by whether the data provide enough information to stabilize the integration step.

4.1 Diagnosing and Investigating the Small-N Problem

After cleaning and pre-processing the data from Mount Pleasant’s 2020 Presidential election, I was left with a sample size of 25 precincts. I input this data into EI to get initial estimates. The analysis was conducted in R using the `ei` package. To run the model, I used the `demo.R` script that comes pre-installed and modified it to work with my dataset, which contains the T, X, and N variables that EI requires. However, this attempt to obtain baseline estimates

produced the following 'integration limits' error:

```
Error in if (any(lower > upper)) stop("lower>upper integration limits") :  
  missing value where TRUE/FALSE needed
```

Troubleshooting this error proved to be difficult: even extensive searches of EI literature and R documentation yielded nearly no discussion of this particular issue.

Looking at the error message, specifically the reference to “integration limits,” it shows that the numerical integration step isn’t converging. Since the model is trying to approximate a posterior distribution, it needs enough data points for that integration to be stable. With such a small number of precincts, there is not enough information in the data to stabilize the likelihood over the feasible parameter space.

A more rigorous breakdown of this error message is as follows:

- **lower > upper:** the model is checking if the integration bounds are valid. These come from the Duncan-Davis constraints in EI. If the computed lower bound is greater than the upper bound, this is mathematically impossible, and the model is not able to run
- **missing values where TRUE/FALSE needed:** at least one of the **lower** or **upper** is missing

In attempt to troubleshoot this error, I looked at the dataset and saw that there were a few extreme points in which X was greater than 0.95 or less than 0.05. These would be precincts where minorities had either extremely high concentrations or low concentrations. Mathematically, this can cause issues with the Duncan-Davis bounds because these bounds require the follow calculations:

$$\frac{T_i}{X_i}, \quad \frac{T_i}{1 - X_i}$$

If $X_i = 0$ or $X_i = 1$, or very close, this can cause a division issue in the above formulas. This would produce an **Inf** or **NA** in EI source code, causing this error message to occur. Especially with small samples, if the model produces this error, bounds become unstable and the integration routine is not able to compute valid limits.

Practically, the turnout assumption imposes that turnout rates are constant across precincts, regardless of demographic composition. When this does not hold, the model may misattribute differences in turnout to differences in vote choice, leading to biased estimates.

With this in mind, I applied the filtering ($X_i > 0.05, X_i < 0.95, N > 50$), and attempted to run the model without extreme precincts. However, this attempt to obtain baseline estimates also failed, again producing the 'integration limits' error. This suggests that the distribution of minority groups among the precincts was not what caused the model to fail.

4.2 Threshold Test of Sample Size

To identify whether there exists a minimum sample size required for EI to successfully run, I conducted a threshold test using Mount Pleasant's 2020 election data. Starting at $N = 25$, I incrementally increased the sample size and recorded whether the model successfully returned estimates at each level. This allows for a direct examination of whether EI failure in small- N settings reflects a consistent threshold or more irregular behavior.

The results of this test with Mount Pleasant's 2020 Election Data were as follows:

N	Result
25	fails
50	fails
51	runs
75	runs

Table 4.1: MP 2020 Threshold Experiment Results

To test whether this was a hard minimum sample size coded into the model itself, this process was repeated using a sample dataset included in King's EI package. However, with this dataset, the model was able to run even with very sparse data ($N < 25$).

Repeating this test with sample data provided within King's EI package showed that the required sample size by the model is not purely dependent on the size of N , but by the distribution of the data, as the sample data converged and ran on sample sizes in which $N < 25$.

N	Result
25	runs

Table 4.2: MP 2024 Threshold Experiment Results

These results show that the minimum sample size threshold depends on the distribution of the data itself, not just the number of observations. In some cases, the model may need more sample data for the numerical integration to converge to estimates. In data-sparse environments with noisy data, it may be the case that EI won't be able to run, and alternative methods of performing statistical analysis should be utilized.

While EI successfully produces estimates for the 2024 dataset, these results should be interpreted with caution. The estimates show only a modest difference (4%) in voting behavior between groups, suggesting limited evidence of strong RPV.

The group voting estimates are as follows:

Group	Voting Rate
Minority	0.5773246
White	0.5345927

Table 4.3: EI Estimates for Group Voting Behavior

However, the deterministic bounds for these estimates add another dimension to this analysis:

Group	Lower Bound	Upper Bound
Minority	0.2484610	0.8432232
White	0.4034653	0.6967713

Table 4.4: EI Bounds for Group Voting Behavior

These bounds are very large, and show that the data, on its own, is not very informative. Because of this, the model has a lot of room to 'choose' estimates, making the estimates strongly susceptible to being impacted by model structure. This shifts the burden of identification from the data to the model assumptions.

However, despite the wide aggregate bounds, EI still produces relatively strong point estimates, which may give a misleading sense of certainty in the model, showing that the

model may be imposing structure where the data provides limited guidance. The model produces tight posterior estimates (small standard deviations) of both the minority and white vote estimates (~ 0.024 and ~ 0.012 , respectively). Given the small-N and the instability in similar settings, like the 2020 data, these results highlight the broader concern that EI can produce seemingly reasonable outputs even when the data is weak.

4.3 Bootstrap Analysis of EI Convergence Across Sample Sizes

To evaluate how sample size affects the likelihood of EI convergence, I conducted a bootstrap analysis on the 2020 election data. For each sample size, precincts were repeatedly resampled with replacement, and EI was estimated on each sample. I define the success rate as the proportion of bootstrapped samples for which the model successfully returned estimates.

The results of this analysis for the 2020 election data is as follows:

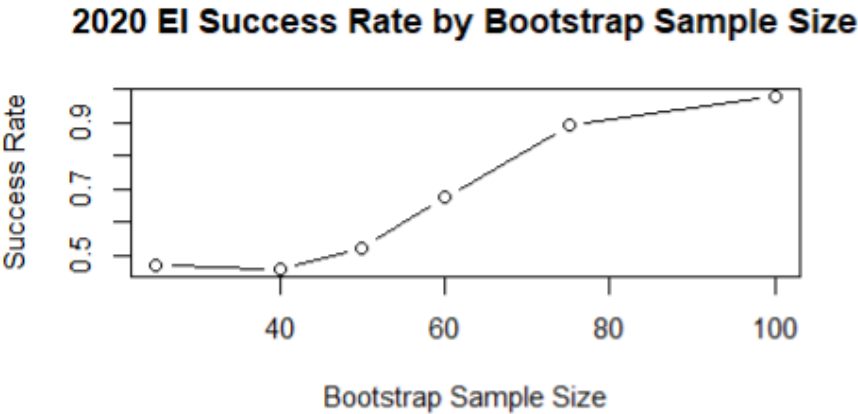


Figure 4.1: 2020 EI Success Rate by Bootstrap Sample Size

Sample Size	Success Rate
25	0.470
40	0.460
50	0.525
60	0.675
75	0.890
100	0.980

Table 4.5: 2020 EI Success Rate by Bootstrap Sample Size

The way the success rate decreases from a sample size of 25 to a sample size of 40 indicates that the data distribution is noisy. If the distribution were more “smooth,” I would expect to see a more stable upward trend as the bootstrap sample size increases.

While a bootstrap sample size of 25 resulted in a success rate of 47%, doubling the sample size led to only a modest increase, to approximately 52.5%. However, tripling the sample size produced a substantial jump in success rate to 89%, and increasing the sample size to 100 resulted in near-perfect convergence. Overall, as the size of the bootstrapped samples increased, the success rate rose as well, demonstrating that increases in sample size improve model stability and convergence.

These findings indicate that both data distribution and small sample size impact the quality and function of EI. In situations where both small-N and an unstable data distribution are present, these factors can interact and amplify one another, worsening their combined impact on model performance.

The results of this analysis for the 2024 election data is as follows:

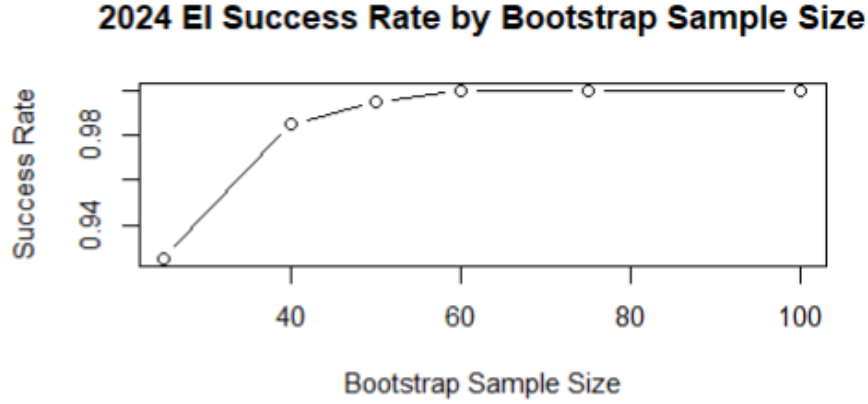


Figure 4.2: 2024 EI Success Rate by Bootstrap Sample Size

Sample Size	Success Rate
25	0.925
40	0.985
50	0.9955
60	1.000
75	1.000
100	1.000

Table 4.6: 2024 EI Success Rate by Bootstrap Sample Size

Compared to the 2020 graph, this graph is smoother and the success rate increases much more rapidly as sample size increases. This indicates that the data distribution for the 2024 data is more stable. While the 2020 data produces an initially convex graph that turns concave, the 2024 graph is entirely concave. This shows the instability of EI in data-sparse environments, as the sample size and success rate of the model have a positive relationship.

Compared to the 2020 test, the success rate for bootstrap samples of size 25 was substantially higher, at approximately 92.5%. Increasing the sample size to 50 resulted in near-perfect convergence, with a success rate of 99.55%. Once the sample size reached 60 and higher, the success rate stabilized at 100%.

The contrast between the elections suggests that the issue is not tied to a specific election, but rather reflects the instability of EI in small-N settings. In both the threshold and bootstrap tests, as sample size increased, the success rate of the model also increased. However, even when datasets have the same number of precincts, differences in the distribution

of demographic composition and vote share can determine whether the model converges. Additionally, the threshold test shows that increasing the amount of data can stabilize the model enough for it to run. As a result, EI does not fail uniformly in small- N contexts and instead behaves inconsistently depending on both sample size and data structure.

4.4 Simulation-Based Analysis of EI Performance in Data-Sparse Environments

To evaluate EI performance in a controlled setting, I conducted a simulation-based analysis using synthetic precinct-level data with known underlying voting behavior. This allows for a direct comparison between estimated and true group voting rates across varying sample sizes.

The results of this experiment are recorded in the following table.

Sample Size	Est. Minority	Est. White	MAE Minority	MAE White
25	0.8999090	0.3002243	0.0010485275	0.0009316484
40	0.8998889	0.3001102	0.0007569459	0.0006737629
50	0.8999007	0.3002350	0.0006536771	0.0006437917
60	0.8997969	0.3002397	0.0006241491	0.0005274737
75	0.8998413	0.3001769	0.0005304339	0.0004724586
100	0.8998207	0.3002344	0.0005061596	0.0004532234

Table 4.7: EI Performance Across Sample Sizes

These results are promising - when the model runs, it performs very well. The estimated minority and white voting rates are extremely close to the true values (90% and 30%, respectively). Additionally, the MAE for both groups is very small, with the largest error still below one-tenth of a percent.

While this analysis didn't utilize Mount Pleasant's election data, these results demonstrate that EI is not inherently unreliable in small samples. When the relationship between demographic composition (X) and vote share (T) is smooth and well-distributed, the model produces accurate estimates even when $N = 25$. The problem is not that small samples

always fail, but that certain data distributions prevent the model’s integration steps from converging. The combination of small- N and unstable data distributions can interact to produce substantial declines in model performance.

4.5 Visualizing the Data Distribution

To better understand the data that converged (Mount Pleasant 2024 Presidential Election & sample EI data) and did not converge (Mount Pleasant 2020 Election data), X on T scatterplots were generated of each of these models. These samples all had uniform $N = 25$. This plots precinct-level demographic composition (X) against vote share (T). The $E(T|X)$ curve represents EI’s estimate of expected vote share given the demographic proportion. The 80% confidence interval shows uncertainty around this relationship.

In EI, this visualization is automated. Below are scatterplots of the data that converged:

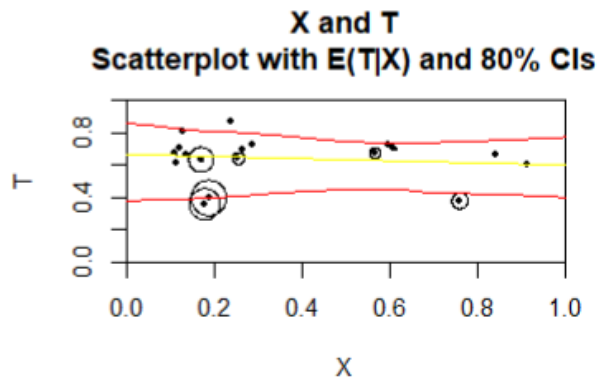


Figure 4.3: X on T Scatterplot for 2024 Presidential Election

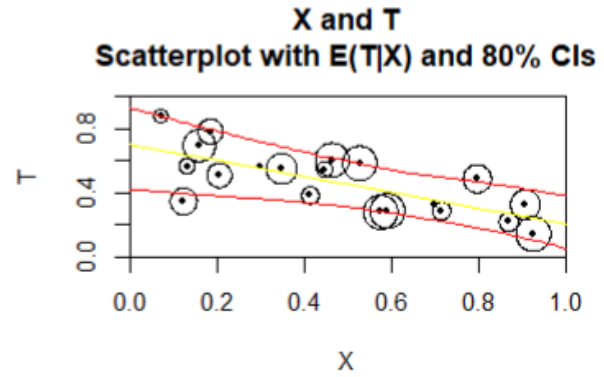


Figure 4.4: X on T Scatterplot for EI Sample Data

- Yellow line: estimated expected vote share given demographic composition
- Red lines: 80% confidence interval
- Points: precincts
- Circle size: how large that precinct is, based on N_i

Below is a scatterplot for Mount Pleasant’s 2020 Presidential election, which did not

converge. As EI failed to run with this data, alternative graphing methods within R were used to visualize this relationship:

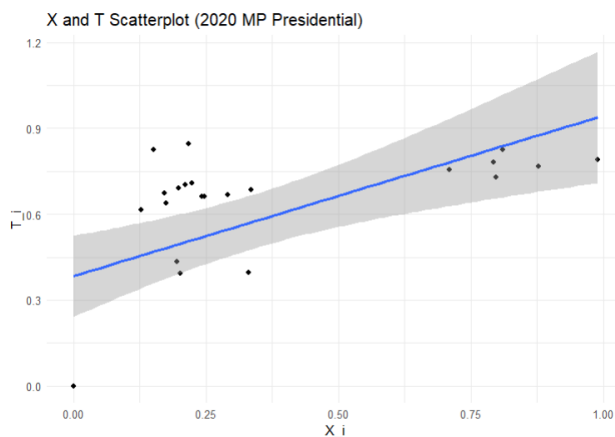


Figure 4.5: X on T Scatterplot for 2020 Presidential Election

- Blue line: estimated expected vote share given demographic composition
- Grey area: 80% confidence interval
- Points: precincts

Visually comparing the plots that converged to the one that did not, the convergent graphs show a more continuous distribution of points along the estimated vote share line. In contrast, the non-convergent graph displays clustered “clouds” of points. This plot has much more noise than the convergent graphs, with a notable extreme outlier at $(0, 0)$. EI software performs best when there is a smooth and informative relationship between X_i and T_i to run and produce estimates. When the relationship between these two points doesn’t have a smooth spread, and the model has less information to learn from. This can cause some bound calculations to produce undefined values and bounds can become unstable. This issue is amplified in small samples, as clustered data further reduces the information available for the model to estimate underlying relationships.

4.6 Mechanism of EI Failure in Small-N and Unstable Data Settings

Taken together, these results show that EI failure is not random, but occurs at a specific step in the estimation procedure. To understand this, I will now examine the mathematical mechanism that is causing this model failure. This section identifies the stage in the EI estimation procedure where failure appears to occur in this application.

EI first uses the tomography line to define all possible combinations of group voting behavior within each precinct. Then it uses the Duncan-Davis bounds to restrict those possibilities to a feasible range. Then, the model defines a likelihood over this range of feasible values and numerically integrates to compute expected values under the assumed distribution. This is where the convergence issue occurs.

The numerical estimation step follows the following procedure:

$$E[\beta] = \int_{\text{lower}}^{\text{upper}} \beta \cdot f(\beta) d\beta$$

Where:

- β : possible value of group voting behavior (either B_i^B or B_i^W)
- $E[\beta]$: Expected value of beta
- $f(\beta)$: how likely each possible value β is, according to the model

In other words, the model computes a weighted average of all feasible values, where weights are determined by the likelihood implied by the truncated bivariate normal assumption.

In order for the model to compute this integral, both the lower and upper bound must be valid, and *lower bound* $\leq$ *upper bound*.

Small sample size combined with an uninformative or irregular data distribution can produce unstable bounds, leading to failure in the numerical integration step.

Chapter 5

Conclusion

By examining the small-N problem in King’s EI, this study demonstrates that both data structure and sparsity contribute to the model’s unpredictability in data-sparse environments. By clearly identifying this limitation, this work lays the groundwork for future research aimed at improving King’s EI so that it can reliably handle data-sparse environments. The lack of attention to the small-N problem presents a challenge for VRA litigation, particularly in geographic regions with a small number of precincts. In these contexts, the model runs unpredictably, limiting its ability to produce consistent and reliable estimates. This makes it difficult to consistently assess the presence of RPV. Because there is no widely accepted software or solution designed to remedy this issue, researchers lack the tools for evaluating RPV in smaller jurisdictions like Mount Pleasant.

When utilizing EI in small-N jurisdictions, researchers are not able to assume that this model will always run or produce stable estimates. Courts that rely on this model should be cautious in data-sparse jurisdictions - this model is context-dependent. Practitioners using EI in legal contexts should evaluate not only sample size, but also the structure of the underlying data before relying on model outputs. Simple fixes don’t perfectly resolve this problem: adding survey data to make the data model more informative introduces bias, and has feasibility issues due to the process of collecting data. Purely using the method of

bounds may be helpful in understanding where group estimates may lie, however, they are often too wide to give clear estimates on group voting behavior. Goodman’s regression has many known methodological flaws that have to be considered when utilizing this method of ecological inference.

This study moves beyond simply noting that EI performs poorly in small samples by identifying why this occurs. By tracing model failure to instability in the integration step caused by the interaction between small-N and data structure, this work provides a more precise understanding of the conditions under which EI can and cannot be relied upon.

This study is subject to several limitations. First, computational constraints limited the number of bootstrap iterations in two of the tests to 200, rather than a larger number such as 1000, which may have provided more stable estimates of model performance. Second, this analysis focuses on a single case study, Mount Pleasant, which may limit the generalizability of these findings to other jurisdictions. Additionally, the simulation framework relies on simplifying assumptions, including fixed turnout and voting rates, which may not fully capture the complexity of actual voting behavior. Finally, this study focuses specifically on EI, and does not evaluate alternative modeling approaches that may perform differently in data-sparse environments.

For future research, one potential direction is incorporating multiple sources of data into EI, as suggested by Wakefield. This could reduce risk when EI is unstable and limits the model’s reliance on non-informative priors. However, this is not always feasible, particularly when studying past elections, and the process of collecting survey results introduces additional assumptions and a potential source of bias. While it’s no longer feasible to apply this approach to the Mount Pleasant data used in this study, it represents a promising direction for future methodological work.

Another potential way to remediate the impact of non-informative priors is by incorporating historical election data into the model to stabilize estimates. Past election data, including Presidential, state, and local races, is available for Mount Pleasant, providing an

opportunity to add additional information to the model. By incorporating this information, the model could “borrow strength” across both precincts and time (Hamra et al., 2013). While this approach has been explored more extensively in the field of epidemiology, it has not yet been examined in the context of RPV.

Using this approach, researchers could construct weakly informative, data-driven priors to inform the expected range of group voting behavior. This would reduce the model’s reliance on non-informative priors and initial assumptions, which make estimates especially unstable in small-N settings. Specifically, prior distributions could be centered around historical estimates of minority and white vote choice, while assigning enough variance to ensure that the model is not forced to replicate past election outcomes. This allows the model to incorporate information from previous elections while still allowing current data to meaningfully influence the final estimates.

More broadly, this paper has shown that EI is not consistently reliable in data-sparse environments where the underlying data are noisy. This was demonstrated empirically through a threshold test, bootstrap analysis of real data, and a bootstrap test using simulated data. Across these tests, I found that EI failure is a function of both sample size and data structure, and behaves inconsistently in small-N settings.

Given the central role of EI in enforcing the VRA, improving its performance in data-sparse environments is critical. While King’s model is widely accepted as the standard for RPV research, it often fails in the contexts where it is most needed. This work highlights that limitation and provides a foundation for future research aimed at improving EI in these settings.

The passage of the Voting Rights Act represents one of the most significant legislative achievements in American history, opening the door to meaningful political representation for communities and groups that had long been excluded from full participation in democracy. Yet, the struggle to ensure equitable representation is ongoing. As new challenges emerge, the tools used to enforce these protections must remain reliable. This paper has shown

that EI, while seen as the standard in VRA enforcement, is not consistently reliable in data-sparse environments where it is needed. If the enforcement of voting rights depends on these methods, then improving their performance is not just a technical concern, but a necessary and important step in protecting the integrity of our democracy. Future research must continue to address these limitations so that promises in the VRA can be fully realized.

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Appendix A

Code Repository

The complete code used for this research is available at the following GitHub repository:

`https://github.com/kdonder1/When-King-s-EI-Fails-Honors-Thesis`

This repository contains all scripts, datasets (if permitted), and instructions necessary to reproduce the results presented in this thesis.